

Fine Tuning a Small Language Model for Sentiment Classification and Evaluating Forgetting on a BoolQ Question-Answering Task

1. Introduction and Problem Statement

Fine-tuning enables a language model to specialize on a particular task, improving in-domain accuracy and reliability. However, this specialization can alter or diminish the model's performance on tasks it previously handled well, a phenomenon known as catastrophic forgetting. Understanding how and why this degradation happens is essential for designing machine learning systems that must operate reliably across multiple domains. As models are increasingly deployed in settings requiring broad generalization, it becomes important to measure not only how well they learn a new task, but also how much they lose in the process.

This project explores these tradeoffs by fine-tuning a compact causal language model on a multiclass sentiment classification task and measuring how this affects its performance on an unrelated yes/no reading-comprehension task modeled after BoolQ. The primary goal is to improve the model's sentiment classification abilities by converting the sentiment dataset into an instruction-style format and training the model to produce the correct sentiment label. The secondary goal is to measure changes in the model's ability to answer yes/no questions grounded in short passages. Evaluations conducted both before and after fine-tuning will reveal how specialization impacts broader reasoning and text-understanding ability. The results will provide insight into the nature of forgetting in small transformer models and emphasize the need for careful monitoring of unintended behavioral changes during fine-tuning.

2. Data Sources

The primary training dataset is the **Sp1786/multiclass-sentiment-analysis-dataset** hosted on Hugging Face. It contains over forty thousand short English text entries labeled as negative, neutral, or positive. Each entry will be transformed into an instruction-style prompt that first presents the text and then asks the model to determine its sentiment. The label will be appended as the expected answer. The training and test splits provided in the dataset will be used without modification, and the entire training portion will be prepared for supervised fine-tuning.

To measure forgetting, the project will use a **larger, automatically constructed BoolQ-style evaluation set** derived from the official [google/boolq](#) dataset. A held-out subset (default size 500) will be sampled and reformatted into an instruction-style structure containing a passage, a yes/no question, and the ground-truth answer. These examples are fully independent from the sentiment dataset and are significantly different in structure and reasoning requirements. None of the BoolQ examples will be included in the fine-tuning data, allowing them to serve as a clean measure of how the model's general comprehension abilities change after training. Both the tokenized sentiment data and the generated BoolQ evaluation file will be stored under the project's [data/](#) directory for reproducibility and consistent loading between notebooks.

3. High-Level Methods, Techniques, and Technologies

The project uses **DistilGPT-2**, a lightweight causal language model designed for efficient training on modest hardware such as CPUs or standard cloud notebook environments. The workflow relies on Hugging Face's **Transformers** and **datasets** libraries for preprocessing, fine-tuning, and evaluation. Sentiment examples are converted into prompt-response pairs by placing the original text under a "Text:" field and appending a question asking for the sentiment, followed by the correct label. Each combined string is converted into a single sequence and tokenized with truncation to a maximum length of 128 tokens. Because GPT-style models do not include a padding token, the EOS token is repurposed as the pad token to reduce warnings and maintain consistent tensor shapes. For causal language modeling, the `labels` field is created as a copy of `input_ids` so that the model learns standard next-token prediction over the combined sequence.

Evaluation is conducted in two stages. First, the base DistilGPT-2 model is evaluated on both sentiment classification and BoolQ to establish baselines. Sentiment evaluation uses greedy decoding with up to three generated tokens, followed by normalization of the output string and comparison to the correct label. BoolQ evaluation uses a log-probability scoring method: for each example, the model computes the likelihood of continuing the prompt with "yes" or "no," and the answer with the higher score is selected as the prediction. After fine-tuning, the same evaluation pipeline is executed again to measure improvements in sentiment classification and changes in BoolQ performance. The difference in BoolQ accuracy before and after fine-tuning is used as the project's primary forgetting metric. All computational steps are executed in Jupyter notebooks using Python, PyTorch, Matplotlib, and Seaborn.

4. Products to Be Delivered

The project will deliver three organized and fully documented Jupyter notebooks, along with a written report and a short video link summarizing the workflow and results. The first notebook prepares the sentiment dataset, converts examples into instruction style prompts, tokenizes the training and test sets, and generates and saves a large BoolQ style evaluation set. The second notebook performs all baseline evaluations, loads or executes the fine tuning process, saves the resulting DistilGPT 2 sentiment fine tuned model, and re-evaluates both sentiment and BoolQ performance to quantify forgetting. The third notebook loads all recorded results, computes metrics, generates visualizations, and provides narrative interpretation of the findings.

Visualizations will include bar charts comparing sentiment accuracy before and after fine tuning, charts comparing BoolQ accuracy before and after fine tuning, a summary bar showing the amount of forgetting, and a confusion matrix for the fine tuned sentiment classifier. Example generations will be included to illustrate qualitative differences in the model's outputs. A structured written analysis in the report will accompany these figures, discussing the motivations for the project, describing the datasets and model, explaining the training pipeline, interpreting all evaluation results, and outlining limitations and possible directions for future work. The notebook set, fine tuned model directory, BoolQ evaluation file, results file, and the video link will serve as a complete and reproducible implementation of the project's objectives.